Jimmy Wang

609-216-4781 | jw4209@columbia.edu

Education

University of California, Berkeley — Berkeley, CA
PhD in Electrical Engineering and Computer Science — *Aug. 2026 –*

- Berkeley Artificial Intelligence Research (BAIR), admitted for Fall 2025 and deferred a year.

Columbia University — New York, NY
Bachelor of Arts in Mathematics. GPA: 3.93 — *Aug. 2021 – May 2025*

- **Selected Graduate Coursework**: Foundations of Optimization (A+), High Dimensional Probability, Real Analysis (A+), Measure Theory, Computational Imaging (A+), Graphics and Computational Motion, Foundations of Graphical Models, Dynamic Programming, Econometrics and Statistical Inference

Works in Progress

1. **Jimmy Wang**, Yifan Chou, Binghao Huang, Kaifeng Zhang, Yixuan Wang, Tao Yu, Saleh Nabi, Yunzhu Li. "Hybrid Particle-Voxel Dynamics for Multimodal 3D World Simulation". Submitted to **ICLR**.

2. Yifan Chou*, **Jimmy Wang***, Yixuan Wang, Xavier Puig, Homanga Bharadhwaj, Yunzhu Li. "Egocentric 3D World Models for Mobile Manipulation Policy Learning.". To be submitted to **RSS**.

3. **Jimmy Wang**, Thomas P. Zollo, Aleksander Holynski, Richard Zemel. "Remembrance (and Reconstruction) of Things Past". To be submitted to **CVPR**.

Publications and Preprints

1. **Jimmy Wang***, Thomas P. Zollo*, Richard Zemel, Hongseok Namkoong. "Adaptive Elicitation of Latent Information Using Natural Language". https://arxiv.org/abs/2504.04204.
ICML 2025
Best Paper at ICLR 2025 Workshop on Quantifying Uncertainty and Hallucination in Foundation Models.

2. Thomas P. Zollo, **Jimmy Wang**, Richard Zemel. "Unsupervised Confidence Calibration for Reasoning LLMs from a Single Generation". https://arxiv.org/abs/2604.19444. *ICML Workshop on Agentic Uncertainty Quantification, 2026*

3. Ethan Che, Daniel Jiang, Hongseok Namkoong, and **Jimmy Wang*** ($\alpha - \beta$). "Optimization-Driven Adaptive Experimentation". *arXiv:2408.04570 [cs.LG].*
Oral Presentation at ESIF Economics and AI+ML Meeting 2024.

4. **Jimmy Wang**, Ethan Che, Daniel Jiang, Hongseok Namkoong. "AExGym: Benchmarks and Environments for Adaptive Experimentation". *arXiv:2408.04531 [cs.LG], 2024.*
Oral Presentation at Conference on Digital Experimentation 2024

5. Jingkai Yan, Shiyu Wang, Xinyu Rain Wei, **Jimmy Wang**, Zsuzsanna Márka, Szabolcs Márka, John Wright. "TpopT: Efficient Trainable Template Optimization on Low-Dimensional Manifolds". https://arxiv.org/pdf/2310.10039.
IEEE Transactions on Signal Processing 2025.

Awards and Honors

- NSF Graduate Research Fellowship 2026
- Best Paper and Oral Presentation at ICLR Workshop on Quantifying Uncertainty and Hallucination in Foundation Models 2025
- Finalist at Undergraduate Informs Prize 2024
- Oral Presentation at Conference on Digital Experimentation 2024

- Oral Presentation at ESIF Economics and AI+ML Meeting 2024
- Columbia Business School Summer Research Award 2024
- Pritzker-Pucker Research Award 2023

Research Experience

Columbia Robotics Lab (RoboPIL) January 2026 - Present

Advised by Yunzhu Li

- First author of paper we have submitted to ICLR, leading industry collaboration with Analog Devices.
- Developing an action-conditioned, 3D world model that can simultaneously output geometry, appearance, and touch in unified 3D space.
- Can simulate scenes interactively under long horizons with challenging in-hand occlusions, while outputting multi-view consistent renders, 3D geometry and tactile particles.

Columbia Computer Science Department January 2026 - Present

Advised by Richard Zemel and Aleksander Holynski

- First author of paper that we are submitting to CVPR.
- Developing 3D reconstruction model that can be queried with a time to output faithful reconstruction of the scene at any chosen moment.
- Our model serves as implicit representation that can be queried using a raymap and time to output appearance and geometry.
- Our model can also be used for active perception, by querying which views have uncertainty and choosing to explore these views.

Zemel Group June 2024 - December 2025

Advised by Richard Zemel

- First author paper that won Best Paper at ICLR Workshop and accepted to ICML.
- Proposed and executed an uncertainty quantification framework for natural language that involves adaptive information gathering, via Monte Carlo Tree Search for information gain maximization.
- Coded entire LLM training and uncertainty quantification pipeline.
- Proved theoretical bounds on optimality of our adaptive information gathering procedure using combinatorial optimization tools.

Columbia Business School Feb. 2023- Jan. 2025

Advised by Hongseok Namkoong and Daniel Jiang

- Two lead author papers that received oral presentations at CODE and ESIF.
- Developed theoretical framework for adaptive experimentation that flexibly incorporates desired objectives and constraints.
- Proved novel sequential central limit theorem result for batched adaptive experimentation.
- Worked with experimentation practitioners at Netflix, Meta, and LinkedIn to construct a general reinforcement learning framework for experimentation.
- Wrote entire codebase and constructed new benchmarks + RL environments for adaptive experimentation.

Columbia Electrical Engineering Department Dec. 2022 - March 2023

Advised by John Wright

- Bounded performance of algorithm using a generic chaining argument on low-dimensional manifolds.
- Co-author of paper accepted to IEEE Transactions on Signal Processing.

Work Experience

Natural Language Processing Intern Jan. 2022 - Mar. 2023

Rights CoLab *New York, NY*

- Led development of Natural Language Processing Pipeline to process 20-F forms to study human/worker rights violations
- Applied sentiment analysis and keyword detection in order to filter database of 10,000+ documents
- Presented findings to Sustainability Accounting Standards Board to make policy recommendations

Talks and Presentations

Optimization-Driven Adaptive Experimentation

- INFORMS Annual Meeting 2024
- Conference on Digital Experimentation 2024
- ESIF Economics and AI+ML Annual Meeting 2024

Adaptive Elicitation of Latent Information Using Natural Language

- ICLR Workshop on Quantifying Uncertainty and Hallucination in Foundation Models 2025

Activities

Columbia Data Science Society Feb. 2023 - May 2025
Executive Board Member *New York, NY*

- Organized yearly hackathons with focus on applications such as patient safety and optimizing subway systems
- Directed and organized panels involving data science and bringing industry leaders